

1. All services running and healthy:

```
veronika.khan@Veronikas-MacBook-Pro gym-app % docker compose ps
NAME                                IMAGE                                COMMAND                                SERVICE    CREATED   STATUS    PORTS
gym-app-auth-service-1             gym-app-auth-service              "java -jar app.jar"                auth-service  18 minutes ago  Up 44 seconds (healthy)    0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp
gym-app-eureka-server-1           gym-app-eureka-server            "java -jar app.jar"                eureka-server  18 minutes ago  Up 49 seconds (healthy)    0.0.0.0:8761->8761/tcp, [::]:8761->8761/tcp
gym-app-gym-core-1                gym-app-gym-core                 "java -jar app.jar"                gym-core       50 seconds ago  Up 38 seconds (healthy)    0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
gym-app-postgres-1                postgres:16-alpine               "docker-entrypoint.s..."          postgres       26 minutes ago  Up 49 seconds (healthy)    0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
gym-app-trainer-report-service-1  gym-app-trainer-report-service   "java -jar app.jar"                trainer-report-service  18 minutes ago  Up 38 seconds (healthy)    0.0.0.0:8081->8081/tcp, [::]:8081->8081/tcp
```

2. Eureka verification:

The screenshot shows the Spring Eureka web interface in a browser. The address bar shows `localhost:8761`. The interface has a dark header with the Spring Eureka logo and navigation links for `HOME` and `LAST 1000 SINCE STARTUP`. A green badge in the top right corner indicates `200` instances.

System Status

Environment	test	Current time	2026-04-15T10:51:31 +0000
Data center	default	Uptime	00:03
		Lease expiration enabled	false
		Renews threshold	6
		Renews (last min)	6

EMERGENCY! EUREKA MAY BE INCORRECTLY CLAIMING INSTANCES ARE UP WHEN THEY'RE NOT. RENEWALS ARE LESSER THAN THRESHOLD AND HENCE THE INSTANCES ARE NOT BEING EXPIRED JUST TO BE SAFE.

DS Replicas

Instances currently registered with Eureka

Application	AMIs	Availability Zones	Status
AUTH-SERVICE	n/a (1)	(1)	UP (1) - 034fa8b3a1d1:auth-service:9000
GYM-CORE	n/a (1)	(1)	UP (1) - e37317c06666:gym-core:8080
TRAINER-REPORT-SERVICE	n/a (1)	(1)	UP (1) - a7b40992d2b9:trainer-report-service:8081

General Info

Name	Value
total-avail-memory	100mb
num-of-cpus	12
current-memory-usage	48mb (47%)
server-uptime	00:03
registered-replicas	
unavailable-replicas	
available-replicas	

Instance Info

Name	Value
ipAddr	172.18.0.2
status	UP

3. Register new trainer via Swagger UI:

The screenshot shows the Swagger UI interface in a web browser at `localhost:8080/swagger-ui/index.html#/Trainers/register`. The interface displays a **POST** endpoint `/api/trainers` with the title "Register trainer".

Parameters: No parameters are listed.

Request body: The body is required and the content type is `application/json`. The JSON body is:

```
{  "firstName": "Victoria",  "lastName": "Benson",  "specializationId": "00000000-0000-0000-0000-000000000002"}
```

Execute: A blue button to execute the request, with a "Clear" button next to it.

Responses:

Curl:

```
curl -X 'POST' \  'http://localhost:8080/api/trainers' \  -H 'accept: */*' \  -H 'Content-Type: application/json' \  -d '{  "firstName": "Victoria",  "lastName": "Benson",  "specializationId": "00000000-0000-0000-0000-000000000002"  }'
```

Request URL: `http://localhost:8080/api/trainers`

Server response:

Code: 201

Response body:

```
{  "username": "Victoria.Benson",  "password": "IFN3i8VodW"}
```

Response headers:

```
cache-control: no-cache, no-store, max-age=0, must-revalidate  connection: keep-alive  content-length: 54  content-type: application/json  date: Wed, 15 Apr 2026 11:00:52 GMT  expires: 0  keep-alive: timeout=60  pragma: no-cache  vary: Origin, Access-Control-Request-Method, Access-Control-Request-Headers  x-content-type-options: nosniff  x-frame-options: DENY  x-transaction-id: ff2d3a8d-c033-4d9f-9111-a27a6dc2d422  x-xss-protection: 0
```

Check in Postgres DB:

The screenshot shows a PostgreSQL database query result in a terminal window. The query is executed against the `gymdb` database on `localhost`. The result is a table with 8 columns: `id`, `first_name`, `last_name`, `username`, `password`, `is_active`, and `failed_login_attempts`. There are 6 rows of data, with the last row highlighted in green.

	id	first_name	last_name	username	password	is_active	failed_login_attempts
1	00000000-0000-0000-0000-000000000010	Alice	Johnson	Alice.Johnson	\$2a\$12\$4Z0nSw4NedTCBt.RHIzeX.5AjALwvNeGI...	true	
2	00000000-0000-0000-0000-000000000011	Bob	Williams	Bob.Williams	\$2a\$12\$SPMgNJ7r5wzV8u0S08cMMOKf9fIM6syF4...	true	
3	00000000-0000-0000-0000-000000000012	Carol	Davis	Carol.Davis	\$2a\$12\$bnnU0nDYxpPnnox9MckV0CoKw3ni2EoJ...	false	
4	00000000-0000-0000-0000-000000000013	John	Smith	John.Smith	\$2a\$12\$fMrEQL6tIHvSfyw9q8QY5.SXUL0SH5SB6...	true	
5	00000000-0000-0000-0000-000000000014	Maria	Garcia	Maria.Garcia	\$2a\$12\$Rup2oPQgi2eLckSw3Nck90sVNxD/nvR7W...	true	
6	f210ac64-01bd-4d90-80f0-3403296ba606	Victoria	Benson	Victoria.Benson	\$2a\$12\$egIzPZQJTCi6w0tK7CDDe01dc87UST7e...	true	

4. Register a new trainee via Swagger UI:

The screenshot shows the Swagger UI interface in a web browser at `localhost:8080/swagger-ui/index.html#/Trainees/register_1`. The interface displays a **POST** endpoint `/api/trainees` for "Register trainee".

Parameters: No parameters are listed.

Request body: The body is required and set to `application/json`. The JSON payload is:

```
{  "firstName": "Kate",  "lastName": "Nordman",  "dateOfBirth": "2001-03-15",  "address": "12 Birch Lane"}
```

Execute: The "Execute" button has been clicked.

Responses: The response section shows a **201** status code. The response body is:

```
{  "username": "Kate.Nordman",  "password": "yGhJXjt8Fz"}
```

The response headers include:

```
cache-control: no-cache, no-store, max-age=0, must-revalidate
connection: keep-alive
content-length: 51
content-type: application/json
date: Wed, 15 Apr 2026 11:13:26 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
vary: Origin, Access-Control-Request-Method, Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: DENY
x-transaction-id: 6eb4944f-bab1-4693-9a31-295e2b4510a0
x-xss-protection: 0
```

Check in Postgres DB:

The screenshot shows a PostgreSQL database query result for the `trainee` table. The query is executed in the `gymdb@localhost [2]` session. The result shows 7 rows of data.

id	first_name	last_name	username	password	is_active	failed_lo
00000000-0000-0000-0000-000000000010	Alice	Johnson	Alice.Johnson	\$2a\$12\$4ZDnSw4NedTCBt.RHizeX.SAJALwvNe6IfRBw9Q8pwMqLIRO8i5J0	true	
00000000-0000-0000-0000-000000000011	Bob	Williams	Bob.Williams	\$2a\$12\$SPMgNJ7rSwzV8u0S08cMMOKf9fIM6syF4Ts86xpVhVxS2srygxxSu	true	
00000000-0000-0000-0000-000000000012	Carol	Davis	Carol.Davis	\$2a\$12\$2bnnU0nDYXpPnnox9McKVOCoKw3ni2EoJQwzr7BTdVzSo1HcR9.G	false	
00000000-0000-0000-0000-000000000013	John	Smith	John.Smith	\$2a\$12\$FMrEQ16tIHvSfyw9q8QYS.SXUL0SHSS86D6Q/hVhXKAyzEAR8nSc6	true	
00000000-0000-0000-0000-000000000014	Maria	Garcia	Maria.Garcia	\$2a\$12\$Rup2oPQgi2e1ckSw3ck90sVNXd/nvR7WwY70q2v/BEisb0EC3lrI	true	
f210ac64-01bd-4d90-80f0-3403296ba606	Victoria	Benson	Victoria.Benson	\$2a\$12\$egIzPZQJTCI66wDtk7CDD0d1dc87UST7eRtJMm0zB.KDX6nmdKxxC	true	
a653cfef-dc00-483d-a45c-0fcee170bee23	Kate	Nordman	Kate.Nordman	\$2a\$12\$CRw.tYssXvBsY6crBCB.X0wHE93xejH7Ahn5VHLyDd0B.RSKhRpCa	true	

4. Login and authorize as newly created trainee and add training:

localhost:8080/swagger-ui/index.html#/Trainings/addTraining

POST

/api/trainings

Add training session

Parameters

No parameters

Request body required

application/json

```
{  "traineeUsername": "Kate.Nordman",  "trainerUsername": "Victoria.Benson",  "trainingName": "Morning Stretch Yoga",  "trainingDate": "2026-04-24",  "duration": 50}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \  'http://localhost:8080/api/trainings' \  -H 'accept: */*' \  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJlYXRlLk5vcnRtYW4iLCJyb2x1IjoiaUk9MRV9UUkF7JTkVFiiwIdWF0IjoxNzc2MjUyNjc1LjE4IjE3NzYzZmZwNzF9.qJQrVSl8opOF2LfusEQVEQFJvewqvJL0xVAKy' \  -H 'Content-Type: application/json' \  -d '{  "traineeUsername": "Kate.Nordman",  "trainerUsername": "Victoria.Benson",  "trainingName": "Morning Stretch Yoga",  "trainingDate": "2026-04-24",  "duration": 50  }'
```

Request URL

http://localhost:8080/api/trainings

Server response

Code

Details

200

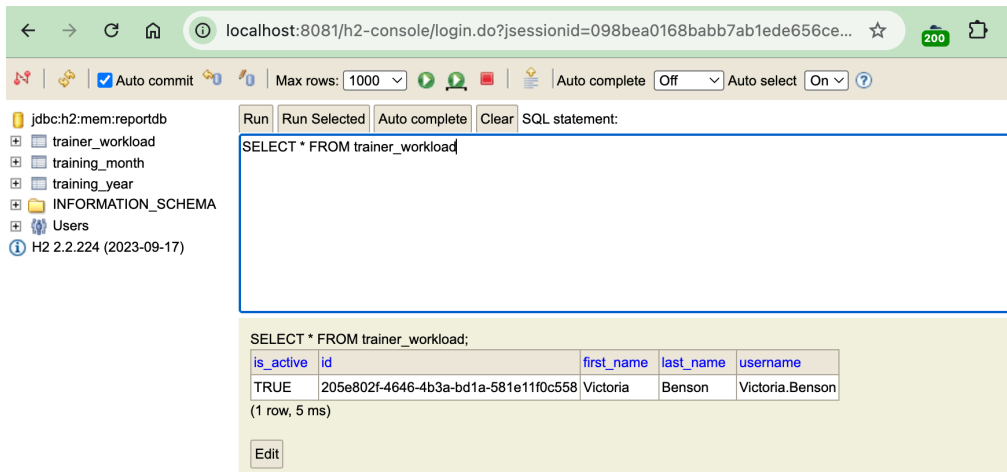
Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate  connection: keep-alive  content-length: 0  date: Wed, 15 Apr 2026 14:33:03 GMT  expires: 0  keep-alive: timeout=60  pragma: no-cache  vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers  x-content-type-options: nosniff  x-frame-options: DENY  x-transaction-id: af631d06-fe59-40ba-908a-77c0bc8a2b4c  x-xss-protection: 0
```

Check Postgres DB:

id	trainee_id	trainer_id	training_type_id	training_name	training_date
00000000-0000-0000-0000-000000000040	00000000-0000-0000-0000-000000000020	00000000-0000-0000-0000-000000000030	00000000-0000-0000-0000-000000000001	Morning Cardio	2025-01-10
00000000-0000-0000-0000-000000000041	00000000-0000-0000-0000-000000000021	00000000-0000-0000-0000-000000000031	00000000-0000-0000-0000-000000000002	Upper Body Stretch	2025-01-12
00000000-0000-0000-0000-000000000042	00000000-0000-0000-0000-000000000022	00000000-0000-0000-0000-000000000031	00000000-0000-0000-0000-000000000002	Full Body Yoga	2025-01-15
7b4df33a-43d2-4f29-bc32-48335e0dc5d4	b2129a72-70e1-452e-81d7-492e5222c269	Sbb9931a-e8e9-4ac2-929c-72d5321765cc	00000000-0000-0000-0000-000000000002	Morning Stretch Yoga	2026-04-20
9b8cfa38-rbf8-4474-8b9d-1454dc22647	b2129a72-70e1-452e-81d7-492e5222c269	Sbb9931a-e8e9-4ac2-929c-72d5321765cc	00000000-0000-0000-0000-000000000002	Morning Stretch Yoga	2026-04-21
6d2d1622-8c27-41ef-975c-3e4e5655a8d59	b2129a72-70e1-452e-81d7-492e5222c269	Sbb9931a-e8e9-4ac2-929c-72d5321765cc	00000000-0000-0000-0000-000000000002	Morning Stretch Yoga	2026-04-22
4c1739ae-6930-43be-8a8d-ac6d11895d4d	b2129a72-70e1-452e-81d7-492e5222c269	Sbb9931a-e8e9-4ac2-929c-72d5321765cc	00000000-0000-0000-0000-000000000002	Morning Stretch Yoga	2026-04-23
60b07210-5176-4e63-808d-a776908c7c9d	b2129a72-70e1-452e-81d7-492e5222c269	Sbb9931a-e8e9-4ac2-929c-72d5321765cc	00000000-0000-0000-0000-000000000002	Morning Stretch Yoga	2026-04-24

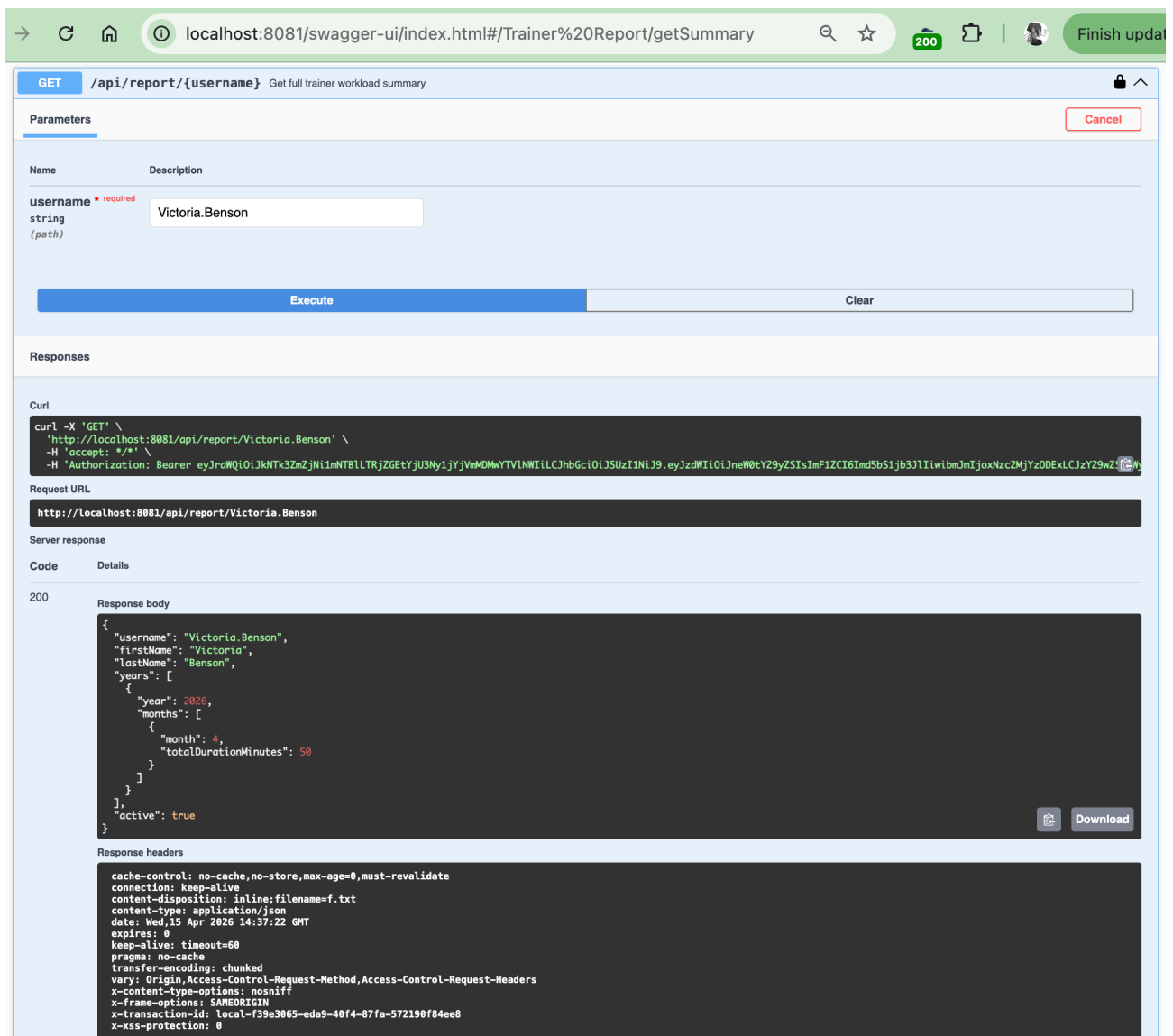
Check H2 DB to verify if trainer-report-service received the information:



The screenshot shows the H2 console interface. On the left, a tree view displays the database structure: jdbc:h2:mem:reportdb, trainer_workload, training_month, training_year, INFORMATION_SCHEMA, Users, and H2 2.2.224 (2023-09-17). The main area contains a SQL statement editor with the query `SELECT * FROM trainer_workload;`. Below the editor, the results are displayed in a table format. The table has five columns: `is_active`, `id`, `first_name`, `last_name`, and `username`. The data row shows `TRUE` for `is_active`, `205e802f-4646-4b3a-bd1a-581e11f0c558` for `id`, `Victoria` for `first_name`, `Benson` for `last_name`, and `Victoria.Benson` for `username`. Below the table, it indicates `(1 row, 5 ms)`. There is an `Edit` button at the bottom left of the results area.

is_active	id	first_name	last_name	username
TRUE	205e802f-4646-4b3a-bd1a-581e11f0c558	Victoria	Benson	Victoria.Benson

Check trainer's workload in Swagger UI (use curl command from [README.md](#) to get access token):



The screenshot shows the Swagger UI interface for the `/api/report/{username}` endpoint. The endpoint is a GET request that returns the full trainer workload summary. The `username` parameter is required and is a string (path). The value `Victoria.Benson` is entered in the input field. The `Execute` button is highlighted. Below the input field, the `Responses` section shows the response for status code 200. The response body is a JSON object containing the trainer's details and workload summary. The response headers are also displayed at the bottom.

Parameters

Name	Description
username * required	
string	
(path)	

Execute **Clear**

Responses

200

Response body

```
{
  "username": "Victoria.Benson",
  "firstName": "Victoria",
  "lastName": "Benson",
  "years": [
    {
      "year": 2026,
      "months": [
        {
          "month": 4,
          "totalDurationMinutes": 50
        }
      ]
    }
  ],
  "active": true
}
```

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
connection: keep-alive
content-disposition: inline;filename=f.txt
content-type: application/json
date: Wed, 15 Apr 2026 14:37:22 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
transfer-encoding: chunked
vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: SAMEORIGIN
x-transaction-id: local-f39e3065-eda9-40f4-87fa-572190f84ee8
x-xss-protection: 0
```

5. Add more trainings for the same trainer and check report via Swagger UI:

The screenshot shows the Swagger UI interface in a web browser. The URL bar indicates the endpoint is `localhost:8081/swagger-ui/index.html#/Trainer%20Report/getSummary`. The interface displays a GET request to `/api/report/{username}` with the parameter `username` set to `Victoria.Benson`. The response status is 200, and the response body is a JSON object containing training data for the specified trainer.

Parameters

Name	Description
username * required string (path)	Victoria.Benson

Responses

200

Response body

```
{
  "username": "Victoria.Benson",
  "firstName": "Victoria",
  "lastName": "Benson",
  "years": [
    {
      "year": 2026,
      "months": [
        {
          "month": 4,
          "totalDurationMinutes": 50
        }
      ]
    },
    {
      "year": 2027,
      "months": [
        {
          "month": 1,
          "totalDurationMinutes": 60
        }
      ]
    }
  ],
  "active": true
}
```

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
connection: keep-alive
content-disposition: inline;filename=f.txt
content-type: application/json
date: Wed, 15 Apr 2026 14:43:34 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
transfer-encoding: chunked
vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: SAMEORIGIN
x-transaction-id: local-1991ccd5-a7ee-4fda-a14b-732e01e852c7
x-xss-protection: 0
```

6. Check trainer's workload for specific months with trainings registered in Swagger UI:

localhost:8081/swagger-ui/index.html#/

200

Finish update

GET

/api/report/{username}/{year}/{month}

Get trainer workload for a specific month

Cancel

Name	Description
username required	
string	Victoria.Benson
(path)	
year required	
integer(int32)	2027
(path)	
month required	
integer(int32)	01
(path)	

Execute

Clear

Responses

Curl

curl -X 'GET' \
'http://localhost:8081/api/report/Victoria.Benson/2027/01' \
-H 'accept: */*' \
-H 'Authorization: Bearer eyJraWQ1OiJkNTk3ZmZlNi1mNTB1LTRjZGEtYjU3Ny1jYjVhMDMwYTU1NWwiLCJhbGciOiJSUzI1NiJ9.eyJzdWIiOiJneW0tY29yZSIsImF1ZCI6Imd5bS1jb3JlIiwibmJmIjojoxNzc2MjY0MjAwLjZyY29wZCJ9'

Request URL

http://localhost:8081/api/report/Victoria.Benson/2027/01

Server response

Code	Details
200	Response body 60 Download Response headers cache-control: no-cache,no-store,max-age=0,must-revalidate connection: keep-alive content-type: application/json date: Wed, 15 Apr 2026 14:44:14 GMT expires: 0 keep-alive: timeout=60 pragma: no-cache transfer-encoding: chunked vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers x-content-type-options: nosniff x-frame-options: SAMEORIGIN x-transaction-id: local-aSab9eb9-0dc6-4b56-8b68-26b7fa0ae2b6 x-xss-protection: 0

Check trainer's workload for the month with no training registered:

→ ↺ 🏠 ⓘ localhost:8081/swagger-ui/index.html#/Trainer%20Report/getMonthlyDurat... 🔍 ☆ 200 📄 👤 Finish update

GET /api/report/{username}/{year}/{month} Get trainer workload for a specific month 🔒 ⌵

Parameters Cancel

Name	Description
username * required string (path)	Victoria.Benson
year * required integer(\$int32) (path)	2027
month * required integer(\$int32) (path)	02

Execute

Clear

Responses

Curl

```
curl -X 'GET' \  
'http://localhost:8081/api/report/Victoria.Benson/2027/02' \  
-H 'accept: */*' \  
-H 'Authorization: Bearer eyJraWQlOiJkNTkzMzZjNi1mNTB1LTRjZGtYjU3Ny1jYjVneMDMwYTVMNWUiLCJhbGciOiJSUzI1NiJ9.eyJzdWIiOiJneW0tY29yZSIImFIZCI6ImdSbS1jb3JlIiwibmJmIjoxNzc2MjY1NTYyLCJzY29wZCJ9.Yn
```

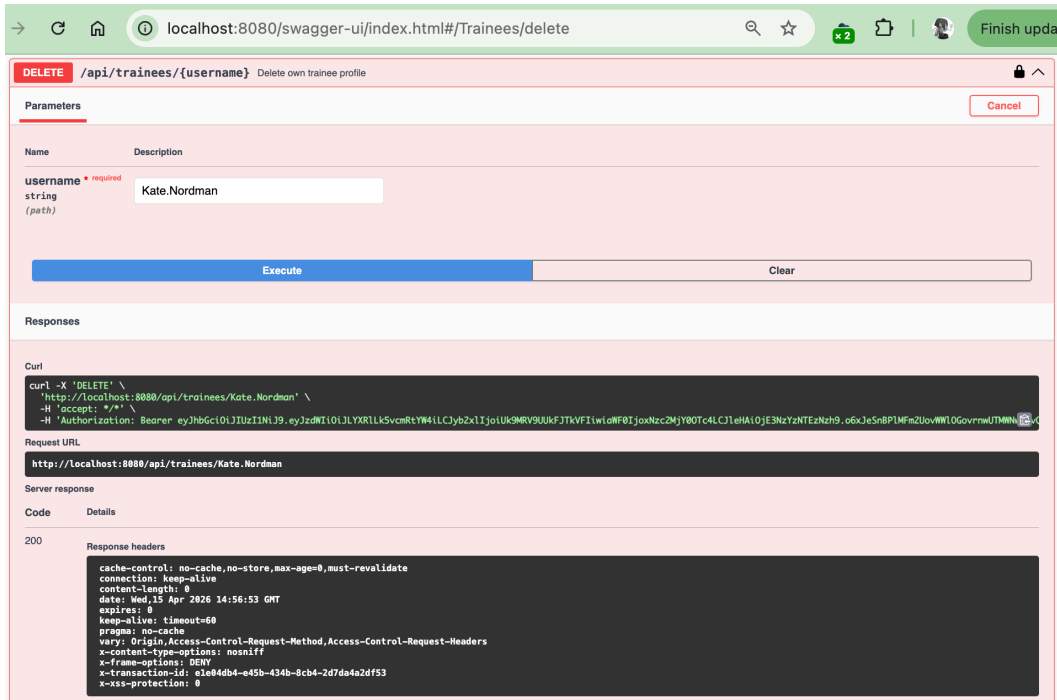
Request URL

```
http://localhost:8081/api/report/Victoria.Benson/2027/02
```

Server response

Code	Details
200	Response body0📄 Download Response headers<pre>cache-control: no-cache,no-store,max-age=0,must-revalidate connection: keep-alive content-type: application/json date: Wed, 15 Apr 2026 15:06:16 GMT expires: 0 keep-alive: timeout=60 pragma: no-cache transfer-encoding: chunked vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers x-content-type-options: nosniff x-frame-options: SAMEORIGIN x-transaction-id: local-312e617f-69d1-4416-ade6-c9a1761e0923 x-xss-protection: 0</pre>

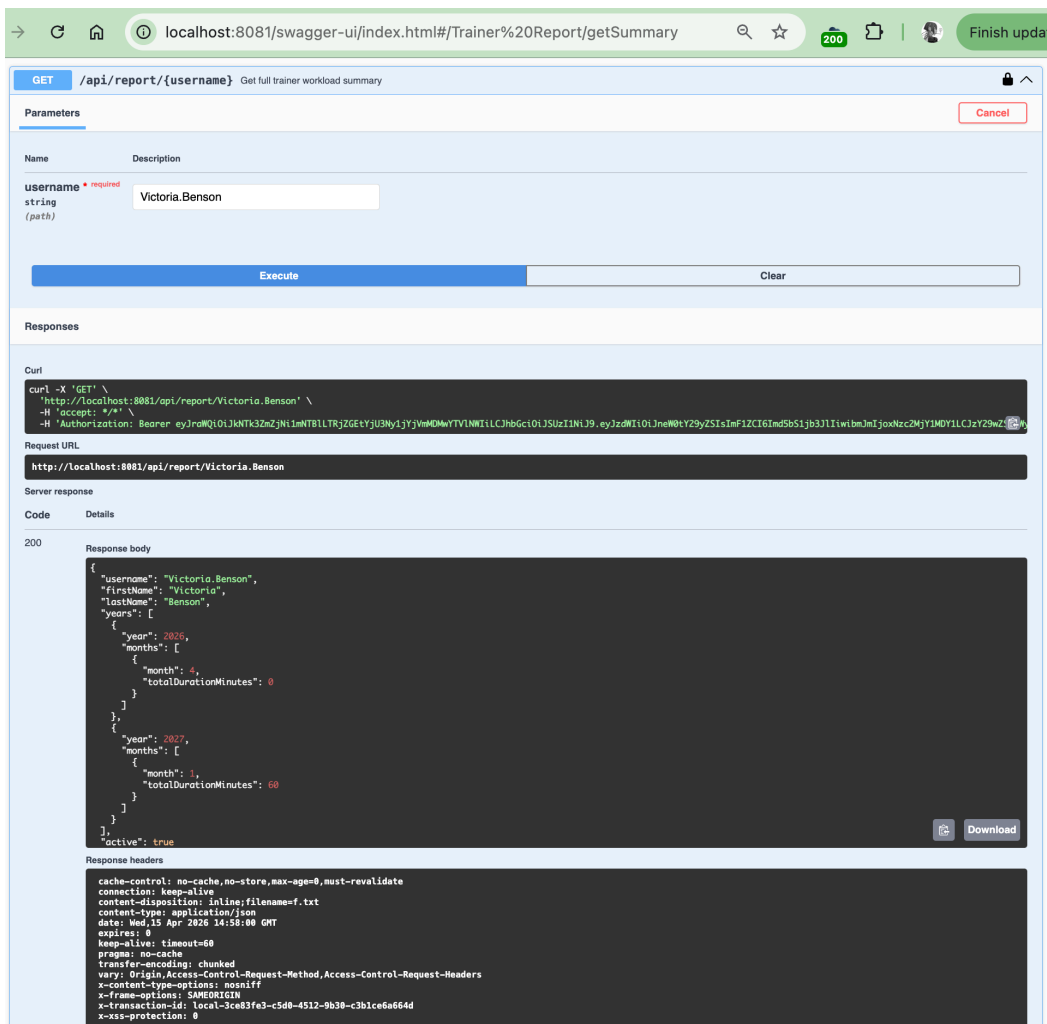
7. Delete one of the trainer's trainees in Swagger UI:



The screenshot shows the Swagger UI for the DELETE endpoint `/api/trainees/{username}`. The `username` parameter is set to `Kate.Nordman`. The response status is 200. The response headers are:

```
cache-control: no-cache, no-store, max-age=0, must-revalidate
connection: keep-alive
content-length: 0
date: Wed, 15 Apr 2026 14:56:53 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
vary: Origin, Access-Control-Request-Method, Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: DENY
x-transaction-id: e1e94db4-e45b-434b-8cb4-2d7da4a2df53
x-xss-protection: 0
```

Check how trainer's workload changed on trainer-report-service side:



The screenshot shows the Swagger UI for the GET endpoint `/api/report/{username}`. The `username` parameter is set to `Victoria.Benson`. The response status is 200. The response body is a JSON object:

```
{
  "username": "Victoria.Benson",
  "firstName": "Victoria",
  "lastName": "Benson",
  "years": [
    {
      "year": 2026,
      "months": [
        {
          "month": 4,
          "totalDurationMinutes": 0
        }
      ]
    },
    {
      "year": 2027,
      "months": [
        {
          "month": 1,
          "totalDurationMinutes": 60
        }
      ]
    }
  ],
  "active": true
}
```

The response headers are:

```
cache-control: no-cache, no-store, max-age=0, must-revalidate
connection: keep-alive
content-disposition: inline; filename=f.txt
content-type: application/json
date: Wed, 15 Apr 2026 14:58:00 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
transfer-encoding: chunked
vary: Origin, Access-Control-Request-Method, Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: SAMEORIGIN
x-transaction-id: local-3ce83fe3-c5d0-4512-9b30-c3b1ce6a664d
x-xss-protection: 0
```

